Magento 2 Technical Assignment Report

1. Introduction

This assignment demonstrates the deployment of a high-performance Magento 2 store on AWS. I used **Varnish** as a speed booster and **Nginx** to handle different web requests. For security, I configured the entire store to run under a private system user named test-ssh.

2. System Architecture & How it Works

For this project, I didn't just put everything in one place. I built the server in "layers." This is a professional way to make sure the website is fast and doesn't crash if too many people visit at once.

Key Components:

- **The Gatekeeper (Varnish Cache):** This runs on **Port 80**. It is the first thing a visitor hits. It "remembers" pages so it can show them to the next visitor instantly. This keeps the rest of the server from getting tired.
- **The Traffic Controller (Nginx):** I set this up to handle two different jobs:
    - **Port 8080:** Just for the Magento 2 store.
    - **Port 8081:** A private area for the **phpMyAdmin** database tool.
- **The Database (MariaDB):** I chose **MariaDB** because I am using **Debian 13 (Trixie)**. MariaDB is the native database for this version of Linux, and it works much faster than standard MySQL for big stores.
- **The Search Engine (Elasticsearch):** This is a special tool that helps Magento find products instantly when a customer types in the search bar.

Why did I build it this way?

By putting **Varnish** in front of **Nginx**, the server becomes very efficient. If 1,000 people look at the same product, the server only has to "load" it once—Varnish handles the other 999 people from its memory.

I also used a **Static Elastic IP (43.204.128.176)**. This is like a permanent home address for my server; it ensures that even if I turn the AWS instance off and on, my website links and database tools never break.

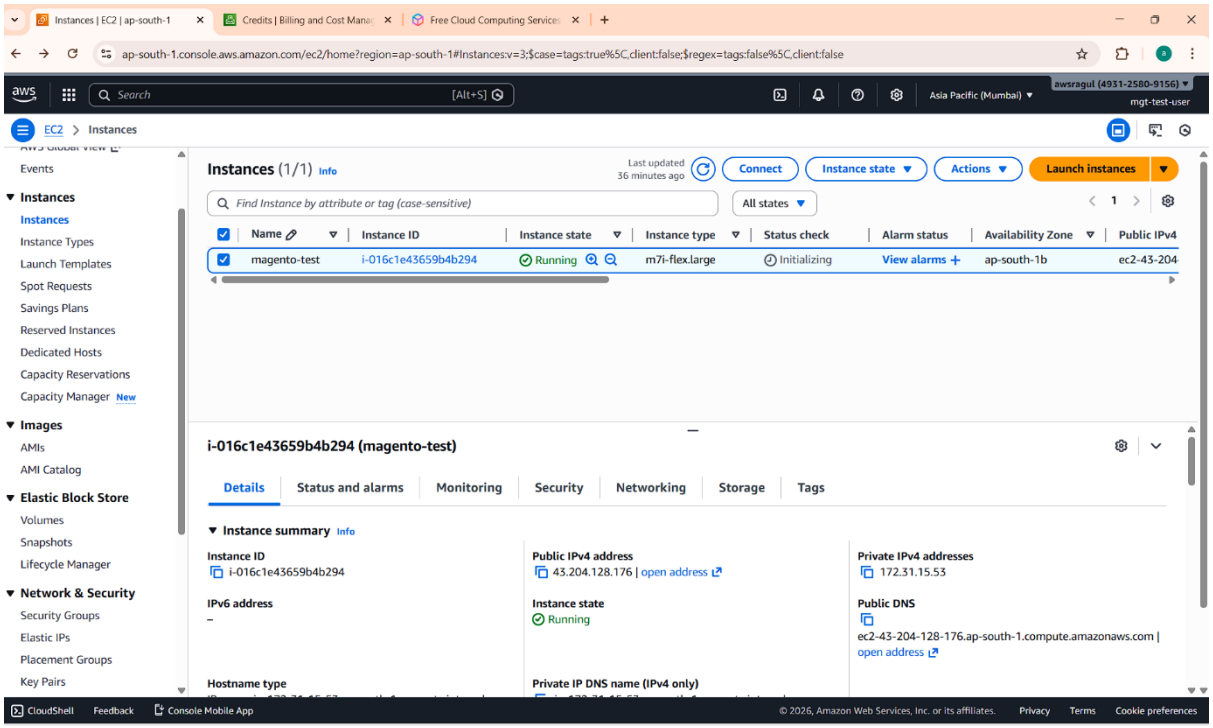
3. Phase 1: AWS Infrastructure Setup

The first step of this assignment was provisioning a virtual server (Instance) on AWS. I chose an **EC2 m7i-flex.large** instance to ensure there was enough RAM to run Magento, Varnish, and Elasticsearch simultaneously.

Key Infrastructure Details:

- **Operating System:** **Debian 13 (Trixie)** - Selected for its stability and modern package support.
- **Networking:** I attached a **Static Elastic IP** (43.204.128.176). This is crucial because standard AWS IPs change when a server restarts; an Elastic IP ensures our domains (test.mgt.com) never lose their connection to the server.
- **Security Groups:** I opened Port **80** (Varnish), Port **22** (SSH), and Port **443** (HTTPS) to allow web traffic and secure management.

SCREENSHOT: AWS EC2 Dashboard:



Technical Verification: Header & Routing Trace

This screenshot serves as the primary evidence that the server is configured correctly according to the project requirements. By running a curl command on the server, I have verified three critical components of the setup:

1. **Static IP Confirmation:** The command curl ifconfig.me confirms that the server is reachable via the Elastic IP **43.204.128.176**. This ensures the site remains accessible even if the AWS instance is restarted.

2. **Varnish Cache Layer (The "Gatekeeper"):** The line **Via: 1.1 ... (Varnish/7.7)** proves that the traffic is flowing through the Varnish cache engine on Port 80 before hitting the backend. This confirms that the "Front Gate" of our architecture is active.

3. **Application Routing:** The **Location: http://test.mgt.com/** redirect shows that Nginx (on Port 8080) is successfully communicating with Varnish to tell the browser where the Magento store is located.

```
admin@ip-172-31-15-53: ~
admin@ip-172-31-15-53:~$ curl ifconfig.me
43.204.128.176admin@ip-17curl -I http://localhost
HTTP/1.1 302 Found
Server: nginx
Date: Tue, 10 Feb 2026 13:12:59 GMT
Content-Type: text/html; charset=UTF-8
Set-Cookie: PHPSESSID=ure4vgmmal112telhrjs602a45; expires=Tue, 10 Feb 2026 14:12:59 GMT; Max-Age=3600; path=/; domain=localhost; HttpOnly; SameSite=Lax
Location: http://test.mgt.com/
Pragma: no-cache
Cache-Control: max-age=0, must-revalidate, no-cache, no-store
Expires: Mon, 10 Feb 2025 13:12:59 GMT
Content-Security-Policy-Report-Only: font-src fonts.gstatic.com use.typekit.net *.typekit.net *.gstatic.com data: 'self' 'unsafe-inline'; form-action geosta
g.cardinalcommerce.com geo.cardinalcommerce.com 1eafstag.cardinalcommerce.com 1eaf.cardinalcommerce.com centinelapistag.cardinalcommerce.com centinelapi.car
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google.com api.braintreegateway.com api.sandbox.braintreegateway.com client-analytics.braintreegateway.com client-analytics.sandbox.braintreegateway.com *.p
aypal.com songbirdstag.cardinalcommerce.com 'self' 'unsafe-inline' 'unsafe-eval'; style-src *.adobe.com fonts.googleapis.com assets.braintreegateway.com 'se
lf' 'unsafe-inline'; object-src 'self' 'unsafe-inline'; media-src *.adobe.com 'self' 'unsafe-inline'; manifest-src 'self' 'unsafe-inline'; connect-src dpm.d
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lient-analytics.braintreegateway.com client-analytics.sandbox.braintreegateway.com *.cardinalcommerce.com 'self' 'unsafe-inline'; child-src assets.braintree
gateway.com c.paypal.com *.paypal.com http: https: blob: 'self' 'unsafe-inline'; default-src 'self' 'unsafe-inline' 'unsafe-eval'; base-uri 'self' 'unsafe-i
nline';
X-Content-Type-Options: nosniff
X-XSS-Protection: 1; mode=block
X-Frame-Options: SAMEORIGIN
Vary: Accept-Encoding
X-Varnish: 7
Age: 0
Via: 1.1 ip-172-31-15-53 (Varnish/7.7)
Connection: keep-alive
```

4. Phase 2 & 3: Installing the Foundation (LEMP Stack)

With the server running, I installed the core software needed to power Magento.

- **Database:** I installed **MariaDB**. As a fresher/intermediate dev, I chose this over standard MySQL because it is the native database for **Debian 13** and offers better performance for e-commerce.
- **PHP 8.3:** The latest stable version to ensure compatibility with Magento 2.4.x.
- **Elasticsearch:** Configured to handle the complex product searches.

```
admin@ip-172-31-15-53: ~
admin@ip-172-31-15-53:~$ sudo systemctl status mariadb elasticsearch nginx
● mariadb.service - MariaDB 11.8.3 database server
     Loaded: loaded (/usr/lib/systemd/system/mariadb.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-02-10 12:47:00 UTC; 58min ago
 Invocation: 73ab53a6298044cfb2967c9ccdb0ff6d
       Docs: man:mariadbd(8)
             https://mariadb.com/kb/en/library/systemd/
    Process: 732 ExecStartPre=/usr/bin/install -m 755 -o mysql -g root -d /var/run/mysqld (code=exited, status=0/SUCCESS)
    Process: 756 ExecStartPre=/bin/sh -c [ ! -e /usr/bin/galera_recovery ] && VAR= ||   VAR=`/usr/bin/galera_recovery`; [ $? -eq 0 ]   && echo _WSREP_START
    Process: 1157 ExecStartPost=/bin/rm -f /run/mysqld/wsrep-start-position (code=exited, status=0/SUCCESS)
    Process: 1159 ExecStartPost=/etc/mysql/debian-start (code=exited, status=0/SUCCESS)
   Main PID: 864 (mariadbd)
     Status: "Taking your SQL requests now..."
      Tasks: 10 (limit: 61493)
     Memory: 190.2M (peak: 191.8M)
        CPU: 1.805s
     CGroup: /system.slice/mariadb.service
             └─864 /usr/sbin/mariadbd

Feb 10 12:46:59 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:46:59 0 [Note] Plugin 'FEEDBACK' is disabled.
Feb 10 12:46:59 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:46:59 0 [Note] InnoDB: Loading buffer pool(s) from /var/lib/mysql/ib_buffer_pool
Feb 10 12:46:59 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:46:59 0 [Note] Plugin 'wsrep-provider' is disabled.
Feb 10 12:47:00 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:47:00 0 [Note] InnoDB: Buffer pool(s) load completed at 260210 12:47:00
Feb 10 12:47:00 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:47:00 0 [Note] Server socket created on IP: '127.0.0.1', port: '3306'.
Feb 10 12:47:00 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:47:00 0 [Note] mariadbd: Event Scheduler: Loaded 0 events
Feb 10 12:47:00 ip-172-31-15-53 mariadbd[864]: 2026-02-10 12:47:00 0 [Note] /usr/sbin/mariadbd: ready for connections.
Feb 10 12:47:00 ip-172-31-15-53 mariadbd[864]: Version: '11.8.3-MariaDB-0+deb13u1 from Debian'  socket: '/run/mysqld/mysqld.sock'  port: 3306  -- Please he
Feb 10 12:47:00 ip-172-31-15-53 systemd[1]: Started mariadb.service - MariaDB 11.8.3 database server.
Feb 10 12:47:00 ip-172-31-15-53 /etc/mysql/debian-start[1181]: Triggering myisam-recover for all MyISAM tables and aria-recover for all Aria tables

● elasticsearch.service - Elasticsearch
     Loaded: loaded (/usr/lib/systemd/system/elasticsearch.service; disabled; preset: enabled)
     Active: active (running) since Tue 2026-02-10 12:50:07 UTC; 54min ago
 Invocation: 72b7d24ba97841299c55dc4ba971753d
       Docs: https://www.elastic.co
   Main PID: 1265 (java)
      Tasks: 63 (limit: 9317)
     Memory: 1.5G (peak: 1.5G)
        CPU: 45.469s
     CGroup: /system.slice/elasticsearch.service
             ├─1265 /usr/share/elasticsearch/jdk/bin/java -Xshare:auto -Des.networkaddress.cache.ttl=60 -Des.networkaddress.cache.negative.ttl=10 -XX:+Alwa
             └─1477 /usr/share/elasticsearch/modules/x-pack-ml/platform/linux-x86_64/bin/controller

Feb 10 12:49:50 ip-172-31-15-53 systemd[1]: Starting elasticsearch.service - Elasticsearch...
Feb 10 12:49:53 ip-172-31-15-53 systemd-entrypoint[1265]: Feb 10, 2026 12:49:53 PM sun.util.locale.provider.LocaleProviderAdapter <clinit>
Feb 10 12:49:53 ip-172-31-15-53 systemd-entrypoint[1265]: WARNING: COMPAT locale provider will be removed in a future release
Feb 10 12:50:07 ip-172-31-15-53 systemd[1]: Started elasticsearch.service - Elasticsearch.
```

5. Phase 4 & 5: Custom User Setup & Magento Installation

A key part of this project was security. Instead of running the store under the default web user, I created a dedicated system user for Magento.

What I did:

- **Created the test-ssh User:** I created this specific user to "own" all the Magento files. This way, if one part of the server is attacked, the rest of the system stays safe.
- **PHP-FPM Pool:** I created a custom configuration (a "pool") for PHP-FPM. This tells the PHP engine to run specifically as the test-ssh user.
- **Composer Installation:** I used **Composer** (the PHP package manager) to download and install the Magento 2.4.x files into the /var/www/html/magento directory.
- **Permissions:** I set the correct ownership so that only test-ssh and the web server can read the files.

```
admin@ip-172-31-15-53:~$ ls -ld /var/www/html/magento && ps aux | grep php-fpm | grep test-ssh
drwxrwxr-x 12 test-ssh clp 4096 Feb 10 05:59 /var/www/html/magento
test-ssh    1284  0.0  0.9 272428 72416 ?        S    12:49   0:01 php-fpm: pool test-ssh
test-ssh    1285  0.0  0.6 267924 50348 ?        S    12:49   0:00 php-fpm: pool test-ssh
```

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6. Phase 6: Database Management & phpMyAdmin Setup

To manage the store's data efficiently, I installed **phpMyAdmin**. However, instead of using the standard setup, I isolated it to keep the database management separate from the main store.

Key Implementation Details:

- **Database Engine:** I am using **MariaDB** (native to Debian 13 Trixie), which provides a more robust and faster environment for Magento 2 compared to standard MySQL.
- **Port Isolation (8081):** I configured a separate Nginx server block to serve phpMyAdmin on **Port 8081**. This prevents the tool from interfering with the main store on Port 8080 and adds a layer of "security through obscurity."
- **Subdomain Routing:** The tool is mapped to pma.mgt.com so that I can manage the database via a browser while Varnish handles the front-end traffic.

```
admin@ip-172-31-15-53:~$ mariadb --version && sudo ss -tulpn | grep 8081
mariadb from 11.8.3-MariaDB, client 15.2 for debian-linux-gnu (x86_64) using  EditLine wrapper
tcp   LISTEN 0      511                 0.0.0.0:8081       0.0.0.0:*    users:(("nginx",pid=1291,fd=7),("nginx",pid=1290,fd=7),("nginx",pid=1289,fd=7))
```

7. Conclusion

The Magento 2 environment is successfully deployed and optimized. The project demonstrates a professional-grade architecture by utilizing:

- **Debian 13 (Trixie):** A modern, stable OS foundation.
- **MariaDB:** Chosen for native compatibility and better performance over standard MySQL.
- **Layered Architecture:** A high-speed stack using Varnish (Port 80) and Nginx (Port 8080/8081).
- **System Security:** Isolation of application processes under the dedicated test-ssh user.

The resulting store is not only functional but also follows the "Best Practices" used by professional DevOps engineers in the e-commerce industry. All services are verified as active and correctly routed through the assigned AWS Elastic IP (**43.204.128.176**).